

Deva Anand

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Education

University of Massachusetts Amherst

Master of Science in Computer Science

September 2025 – May 2027 (Expected)

Amherst, MA

Coursework: Performance Engineering, Systems for Data Science, Optimization in CS, Advanced Machine Learning, Advanced NLP

Vels Institute of Science, Technology and Advanced Studies

Bachelor of Engineering in Computer Science and Engineering

August 2019 – May 2023

Chennai, India

Skills

Languages

C++, Python, JavaScript, TypeScript, Java, SQL, Bash

Systems & Performance

Linux profiling & benchmarking, multithreading, OpenMP / SIMD concepts, memory-hierarchy / cache-aware design, performance experiments, Spark / PySpark, Spark UI

Testing & CI/CD

pytest, GTest / CMake, integration & regression testing, test automation, Git / GitHub, GitHub Actions, CI/CD, Agile / Scrum

Backend

Node.js, Express, FastAPI, REST APIs, async services, multi-tenant SaaS

Cloud & DevOps

AWS, GCP, Azure, Docker

Databases

MongoDB, PostgreSQL, Redis, SQLite

Experience

Wysa

Backend Engineer

July 2023 – July 2025

Bengaluru, India

- Took features through the full software lifecycle (requirements to design, implementation, verification, and release) on **multi-tenant NHS eTriage systems** (Node.js, MongoDB) serving **8+ UK clients** and **10,000+ monthly submissions** integrated with Mayden's iaptus platform; contributed to **\$340K+** in revenue.
- Monitored production logs during releases and client integrations, investigated incidents, and reproduced and shipped fixes for tenant-specific and integration issues in an **Agile/Scrum** team (sprint planning, stand-ups, retrospectives).
- Built **integration and regression tests** and remediated **20+ VAPT findings** to protect critical workflows against regressions; ran knowledge transfer and design walkthroughs to onboard a second developer to on-call readiness.

Cario Growth Services Pvt. Ltd.

Backend Developer Intern

November 2022 – May 2023

Chennai, India

- Designed and implemented **20+ RESTful APIs** in Node.js + Fastify backed by **PostgreSQL**, with input validation, clear contracts, and backward-compatible changes used in production by the frontend team.

Projects

Spark ETL Performance Optimization (NYC TLC Trip Data)

github.com/Deva-1903/Spark-ETL-Optimization

Apache Spark / PySpark, Spark UI, SQL, Python

- Ran controlled performance experiments on a **~1.4B-row** dataset, cutting end-to-end ETL runtime **~25%** (17–19 min to 14.3 min) by tuning broadcast joins, AQE, skew-join handling, shuffle partitions, and partitioning, and shared the results as a repeatable optimization playbook.
- Profiled stage-level timings via **Spark UI** to localize shuffle and skew bottlenecks, reducing analytics tail-latency ratio (P99/P50) from **2.14x to 1.74x** (~19%) through column pruning, filter pushdown, and precomputed fields.

Sonare - Offline Sign ↔ Speech Cross-Platform Desktop App

github.com/Deva-1903/Qualcomm-Sep25-Team-Sonare

React, Electron, FastAPI, MediaPipe, whisper.cpp | Qualcomm Edge AI Hackathon 2025

- Built a cross-platform desktop app integrating camera, microphone, UI, and backend into an **on-device pipeline** (MediaPipe hand tracking, whisper.cpp ASR, local TTS), bridging the UI and local processing layers under hackathon time pressure.
- Designed async APIs and queues to keep the UI responsive while handling local video and audio streams; ran **fully offline** with no external network dependencies.

AgenticSearch - Provenance-First Entity Discovery

github.com/Deva-1903/ciir_agentic_search

Python, FastAPI, pytest, OpenAI / Groq, SQLite

- Hardened a multi-stage service with **150+ automated tests** and provider routing/fallback, plus pipeline instrumentation for review-friendly debuggability and regression protection across backends.

Publication & Extracurriculars

- **Publication:** "Alzheimer's Disease Classification using Transfer Learning," **IEEE CONIT 2023**.
- Selected for **GirlScript Summer of Code (GSSoC) 2023**; contributed to multiple open-source projects (Linkfree, Freehit, ProjectsHut).
- Tutored **30+** underprivileged students in programming at Sayur, non-profit in South Tamil Nadu.